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Ras-mediated dynamic and biphasic regulation of cell migration

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Ras has traditionally been regarded as a positive regulator and therapeutic target due to its role in cell proliferation, but recent findings indicate a more nuanced role in cell migration, where suppressed Ras activity can unexpectedly promote migration. To clarify this complexity, we systematically modulate Ras activity using various RasGEF and RasGAP proteins and assess their effects on migration dynamics. Leveraging optogenetics, we assess the immediate, nontranscriptional effects of Ras signaling on migration. Local RasGEF recruitment to the plasma membrane induces protrusions and new fronts to effectively guide migration, even in the absence of GPCR/G-protein signaling, whereas global recruitment causes immediate cell spreading halting cell migration. Local RasGAP recruitment suppresses protrusions, generates new backs, and repels cells, whereas global relocation either eliminates all protrusions to inhibit migration or preserves a single protrusion to maintain polarity. Consistent local and global increases or decreases in signal transduction and cytoskeletal activities accompany these morphological changes. Additionally, we performed cortical tension measurements and found that Ras activity is regulated by guanine nucleotide exchange factors generally increase cortical tension while Ras activity is regulated by GTPase-activating proteins decrease it. Our results reveal a biphasic relationship between Ras activity and cellular dynamics, reinforcing our previous findings that optimal Ras activity and cortical tension are critical for efficient migration.

cell migration | Ras GTPase | cortical tension

Significance

This study challenges the traditional view of Ras as only a positive regulator of cell functions via gene expression. Using optogenetics to rapidly modulate Ras activity in *Dictyostelium*, we reveal a biphasic relationship: both excessive and insufficient Ras activity rapidly impair cell migration, independent of transcriptional change. These findings show that optimal Ras activity and cortical tension are essential for efficient migration. Importantly, targeting Ras in cancer therapy should consider the cell's initial state, aiming to shift Ras activity outside the optimal range for migration. This nuanced understanding implies that simply inhibiting Ras could inadvertently promote metastasis, highlighting the need for more sophisticated approaches to targeting Ras for effective cancer treatment.

Ras GTPases are highly versatile proteins that play pivotal roles in numerous signaling events, regulating fundamental cell functions such as growth, survival, differentiation, and migration (1–4). A delicate balance of Ras activity is crucial for proper gene expression, cellular functions, and tissue development, and dysregulation of Ras can lead to developmental abnormalities and the onset of diseases such as cancer (5–7). It is generally believed that Ras serves as a *positive* regulator of these functions, and Ras has garnered enormous attention as a therapeutic target, with most research focused on proliferation and differentiation (8–10). However, recent studies of cell migration have shown that effects of modulating Ras activity depend largely on the initial state of the cell. Studies in rapidly moving cells, such as *Dictyostelium discoideum* and human neutrophils, have shown that constitutively active Ras isoforms induce more protrusion formation as might be expected, but excess protrusions can lead to cell spreading that precludes movement. Interestingly, dominant negative Ras isoforms or the regulation of Ras activity by GTPase-activating proteins (RasGAPs), that limit protrusions, can promote cell polarization and migration (11, 12). These findings highlight the complexity of Ras signaling in cell migration and show that the relationship between Ras activity and cell migration is not straightforward. The counterintuitive outcomes must be considered when assessing Ras as a target to treat cancer. An inhibitor that limits growth may in fact promote metastasis. Therefore, it is important to obtain a full picture of how modulating Ras activity affects cell migration, as well as other aspects, such as cell morphology, macropinocytosis, and mechanical properties.

During growth factor stimulation, transcriptional and translational events are not immediately observable, whereas the triggering of Ras/Raf/MEK/ERK and Ras/PI3K/mTOR signal transduction cascades is readily apparent. Cells undergo immediate morphological changes caused by dramatic cytoskeleton remodeling, leading to cell spreading, ruffles, and protrusion formation (13–16). Furthermore, optogenetically activating or inhibiting Ras can rapidly create new protrusions and guide cell migration or eliminate protrusions and halt migration, respectively, on timescales much faster than those required for changes in gene expression (11, 12, 17, 18).

Mutations in the human NF1 gene, a RasGAP, cause the severe genetic disorder neurofibromatosis type 1 (19, 20). The defects are typically associated with increased

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proliferation with little consideration of effects on cell morphology and dynamics (21, 22). However, macropinocytosis is upregulated in Ras-driven tumors and the process appears to be essential for driving accelerated growth. Interestingly, activation of Ras isoforms is dynamically associated with macropinocytotic cups in axenic strains of the social amoeba *Dictyostelium* (23). This upregulation of macropinocytosis in these strains is caused by inactivation of a single NF1 homologue which leads to hyperactivated Ras activity and underlies the massive increase in macropinocytosis. Surprisingly, cells with activated Ras do not grow dramatically faster, suggesting that G1 checkpoint control is not critical in these free-living amoebae. For these reasons, this model organism offers the possibility of studying the effects of Ras activity modulation on cell dynamics without complication from effects on proliferation.

Ras activity is regulated by guanine nucleotide exchange factors (RasGEFs) and RasGAPs (24, 25). The *Dictyostelium* genome includes about 25 genes encoding RasGEFs and 14 for RasGAPs (26). Among them, Aimless and GefR can activate RasC and RasG, and their loss-of-function mutants have defects in chemotaxis (27). GefB, GefM, GefU, and GefX are all reportedly important for random migration and spontaneous signaling events (26). RasGAP, C2GAP1, interacts with RasC and RasG and is essential for GPCR-mediated Ras signaling in chemotaxis (28, 29). RasGAP, C2GAP2, is the common regulator of Ras signing in GPCR-mediated chemotaxis, macropinocytosis, and phagocytosis (30). NF1, NfaA, IqgC, Leep2A, and Leep2B are RasGAPs that are mainly involved in the regulation of macropinocytosis or phagocytosis (23, 31–34). Previously, we found that overexpression or optogenetic recruitment of C2GAPB in vegetative *Dictyostelium* unexpectedly polarized cells, albeit its inhibitory role on Ras activity (11). Considering the complexity and importance of Ras signaling on cell shape and migration, we aim to systematically modulate Ras activity over a broad spectrum and further extend our investigation to GefA (Aimless) and NGAP (C2GAP1). Through overexpression or optogenetic manipulation, we sought to determine their effects on cell migration, cell morphology, signal transduction, macropinocytosis, as well as the mechanical properties of cells. Our evidence suggests a biphasic relationship between Ras activity and cell migration. By dissecting this critical node in the signaling network, we provide insights into the coordinated regulation of cellular processes by Ras signaling pathways.

Results

Exchange Factor GefA Alters Cell Morphology and Migration. Because Ras is known to function as a positive regulator in cell migration and chemotaxis (12), we first examined the immediate effects of increased Ras activity on cellular protrusions and migration directions. To accomplish this aim, we designed an optogenetic system to recruit SspB fused GefA protein from the cytosol to the cell membrane-anchored iLiD when blue light is presented (35–37). We hypothesized that recruiting more GefA protein at the front of cells would enlarge the protrusions, while recruiting it to the back would create new protrusions and reverse migration directions (Fig. 1*A*). When we locally illuminated a protrusion, the recruitment led to an expansion of the protrusion within 30 s and induced cells to shift the migration direction (Fig. 1*B*). Next, we attempted to make the cell reverse direction by recruiting GefA to the back. This manipulation did induce protrusions, but because vegetative *Dictyostelium* cells are less polarized, the direction changes were inconclusive. Therefore, we loaded cells into microchannel devices which enforce a defined direction. As shown in Fig. 1*C*, when GefA-expressing cells moved in one direction, we locally concentrated GefA at the trailing edge. In less than 3 min, the cell switched direction. We were able to repeatedly change directions by alternately recruiting GefA at different positions. The movement of the edge of the cell lagged the recruitment of the GefA by 20 to 60 s (Movie S1). Compared to no light treatment where cells move to either left or right, the recruitment of GefA consistently guided cells move toward the side of GefA accumulation (Fig. 1*D*). We also validated this result with empty vector control (opto-Ctrl) recruitment, which did not alter cell migration direction as indicated by the positive value of velocity before and after laser application (*SI Appendix*, Fig. S1*A*). On the contrary, opto-GefA recruitment changed the velocity to negative value, indicating a directional change (*SI Appendix*, Fig. S1*B*). Strikingly, local recruitment of GefA can induce new protrusions and guide cell migration in Gβ-null cells, which are defective in most receptor-mediated signal transduction events and do not carry out chemotaxis (*SI Appendix*, Fig. S1*C*).

Given the obvious effects of increasing Ras activity on promoting cellular protrusions and guiding cell migration direction, we hypothesized that overexpressing GefA would promote cell migration. Suspecting that inhibitory effects of continuous expression of active Ras proteins on cytokinesis selects for low expressors, we used an inducible system to control GefA expression. Axenic vegetative cells exhibit an amoeboid shape and often form several protrusions simultaneously. They are less polarized then developed cells, but nevertheless display moderate movement. However, upon the overnight induction of GefA expression, the cells exhibited either a large, flat, pancake-like single global protrusion or thin, elongated morphologies with multiple opposing protrusions, depending on the expression level of the GefA (Fig. 1*E*). In either case, a major reduction in migration speed and macropinocytosis occurred (*SI Appendix*, Fig. S1 *D* and *E*). Time-lapse videos of GefA-expressing cells revealed that both morphological categories of cells barely move with almost no changes in cell shape. The cells appeared "frozen" over 10 min, compared to the control cells which displayed frequent shape changes and movement. The differences in migration are highlighted by the color-coded overlay (Fig. 1*E*). Because of the flattened or stretched-out morphology, the average height of the GefA-expressing cells was around half of the control (Fig. 1*F*), and the cell basal area and flatness increased more than twofold (Fig. 1*G* and *SI Appendix*, Fig. S2*A*). Quantification of cell migration speed showed that GefA-expressing cells on average migrated at 0.5 to 1 μm/min, while wild-type cells migrated at 6 μm/min (Fig. 1*H*). The average aspect ratio of GefA-expressing cells and wild-type cells was around 1.5, while that of the elongated fraction ranged from 2.5 to 4.5 (*SI Appendix*, Fig. S2*B*). Because the GefA-expressing cells have two different morphologies, we suspected that it could be the heterogeneous expression differences. Consistently, as shown in *SI Appendix*, Fig. S2*C*, among the GefA-expressing cells, the higher expression levels of GefA have a larger basal area and smaller aspect ratio.

To confirm that this phenotype was indeed caused by GefA rather than a secondary effect of its expression, we conducted global recruitment of GefA. We anticipated that global recruitment would not only recapitulate the overexpression phenotype but also produce a stronger effect due to its faster and more direct activation. As shown in Fig. 1*I*, global recruitment of GefA rapidly induced multiple protrusions and expanded the cell in every direction, resulting in increased basal area and immobilization (Movie S2), as quantified in Fig. 1 *J* and *K*. The aspect ratio did not change significantly due to the irregular, but symmetrical, shape of GefA-expressing cells before recruitment (*SI Appendix*, Fig. S2*D*). The color-coded overlay illustrates the changes in movement and

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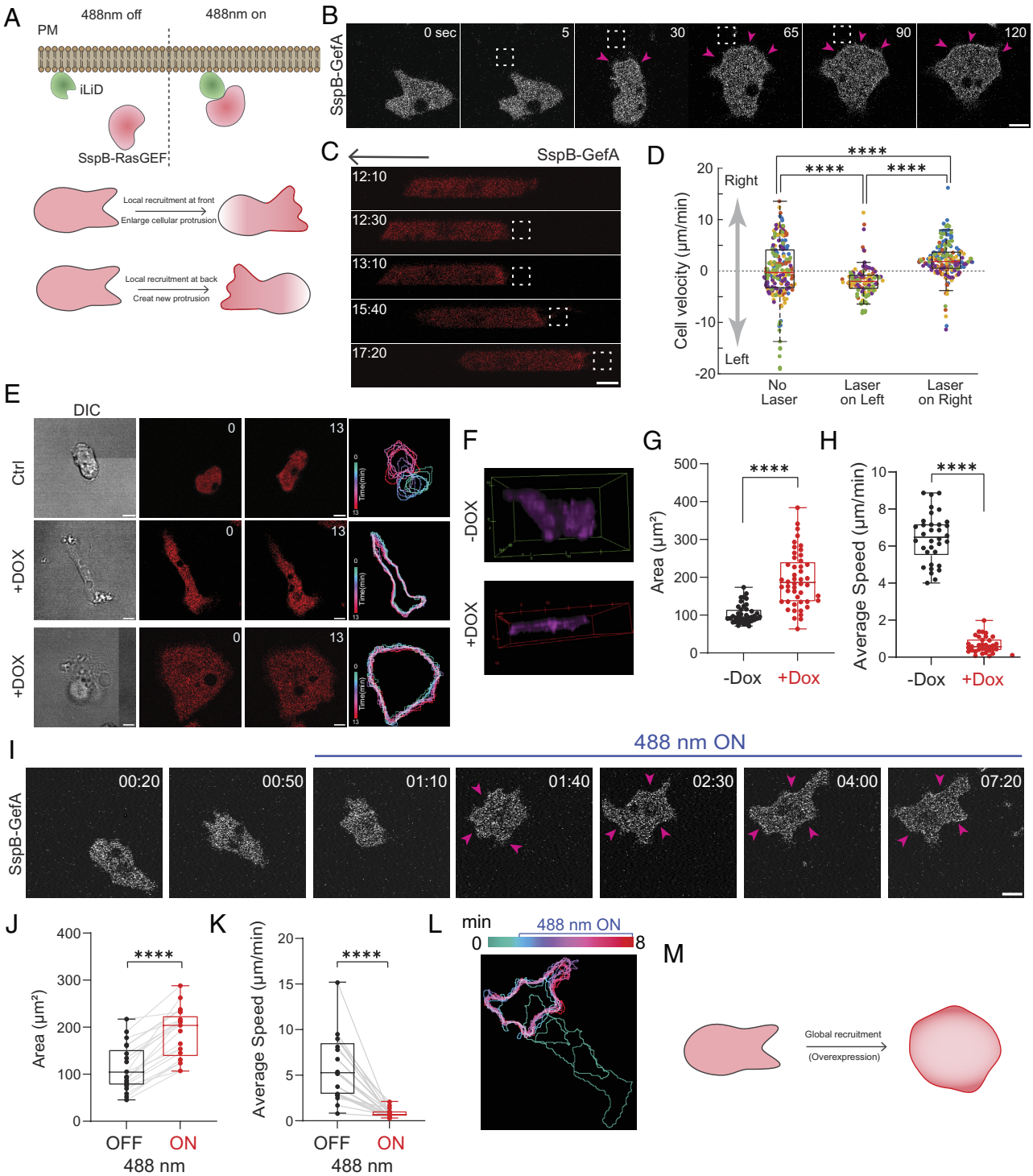

Fig. 1. Activation of Ras by GefA alters cell morphology and migration. (*A*) Schematic of the optogenetic tool. Recruitment of Opto-GefA (mRFPmars-SspB-GefA) to membrane-anchored iLiD upon 488 nm illumination. Typical effects at cell front or back are indicated. (*B*) Opto-GefA recruited near a protrusion from migrating vegetative *Dictyostelium* cells on the 2D glass surface with 488 nm light (dashed white box). Pink arrows highlight the expansion of the protrusion. (Scale bars, 5 μm.) (*C*) Opto-GefA recruited to the back of the cell migrating to the left in a microchannel (dashed white box). The arrow indicates the migration direction before light stimulus. (Scale bars, 5 μm.) (*D*) Instantaneous velocity of Opto-GefA cells when the laser light is off, to the left or right of cells. Colored dots represent each of n = 5 cells; total number of measurements are 215, 121, and 184, respectively. Asterisks indicate significant difference, ****$P \leq 0.0001$ (two-sided Mann–Whitney test). (*E*) Time-lapse confocal images of vegetative *Dictyostelium* cells expressing empty vector control (red; *Top*) or mRFPmars-GefA (red; *Middle* and *Bottom*) after overnight doxycycline (Dox) treatment. Time in minutes. (Scale bars, 5 μm.) DIC images (*Left-most* panel) and overlaid outlines (color-coded 1-min intervals) of control or GefA-expressing cells (*Right-most* panel). (*F*) 3D z-stack images of GefA-expressing cell minus (*Top*) or plus (*Bottom*) Dox induction. Box-and-whisker plots of cell area (*G*) and average cell speed (*H*) before (black; −DOX) or after (red; +DOX) overnight induction of GefA. n = 40 (−Dox) or n = 49 (+Dox) cells examined over three independent experiments; asterisks indicate significant difference, ****$P \leq 0.0001$ (two-sided Mann–Whitney test). The boxes extend from 25th to 75th percentiles, the median is at the center, and whiskers and outliers are graphed according to Tukey's convention (GraphPad Prism 8). (*I*) Snapshots of vegetative *Dictyostelium* cells expressing Opto-GefA on the 2D glass surface. Opto-GefA is recruited globally by switching on a 488 nm laser. Pink arrows highlight the expansion of the entire cell. Time in min:s format. (Scale bars, 5 μm.) Box-and-whisker plots of cell area (*J*) and average cell speed (*K*) before (black; 488 nm OFF) or after (red; 488 nm ON) global recruitment of GefA. n = 21 cells examined over three independent experiments. Statistical analysis as in (*G* and *H*). (*L*) Color-coded (1-min interval) outlines of a representative cell before and after the presence of 488 nm light. (*M*) Cartoon summarizes the effects of global opto-GefA recruitment or overexpression of GefA.

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cell shape before and after GefA recruitment, showing that after recruitment, the cells expanded with more protrusions in all directions, resulting in zero net movement (Fig. 1*L*). Similarly, the global recruitment of GefA expanded the Gβ-null cells in all directions, leading to an impairment in migration, and captured by color-coded images (*SI Appendix*, Fig. S2 *E* and *F*). Taken together, overexpressing GefA or global recruitment of GefA that increased Ras activity resulted in cell expansion and impaired cell movement, as summarized in Fig. 1*M*.

Exchange Factor GefA Activates Signal Transduction and Cytoskeletal Activities. Since overexpression or recruitment of GefA led to remarkable changes in cell morphology and alterations in migration, we investigated whether the phenotypes were indeed caused by hyperactivated Ras activity. We compared Ras activity in cells with and without GefA expression using fluorescence-labeled RBD as a biosensor. As shown in Fig. 2*A*, control cells showed micron-sized, dynamic patches of RBD signals that marked protrusions, typically lasting for 1 or 2 min. In contrast, GefA-expressing cells displayed much larger RBD patches on the membrane that remained relatively static for about 10 min in this example. When normalized to the cell perimeters, the lengths of the RBD patches are increased more than twofold in the GefA-expressing cells compared to controls (Fig. 2*B*). Occasionally, RBD waves were observed in single GefA-expressing cells, propagating across the cell and creating a broad protrusion as the wave swept to the edge (Fig. 2*A*). Such waves of Ras activity are observed in electroshock-fused giant cells but are rare in single control cells not expressing GefA. To further assess whether GefA increases events downstream of Ras activity, we used biosensor PHcrac and LimE to examine PIP3 and F-actin polymerization activity. Normally, PHcrac and LimE mirror RBD activity, occurring with protrusions, as shown in the controls (Fig. 2*C*). As expected, in GefA-expressing cells, LimE mimicked the RBD signals and indicated significantly increased actin polymerization along the cell periphery, which could account for their flattened phenotype. However, we did not observe PHcrac labeling on GefA-expressing cell membranes (Fig. 2*D*). Additionally, to confirm that GefA expression promotes actin polymerization, we fixed cells and stained them with phalloidin, an F-actin binding reagent. As shown in Fig. 2*E*, phalloidin mainly stained macropinocytotic cups in control cells, but in GefA-expressing cells, it exhibited massive staining on the entire cell membrane along with many diffuse, short filament structures in the cytosol. These results together indicate that GefA expression increases Ras activity and promotes more actin polymerization around the entire cell periphery, which likely explains the decreased migration in GefA-expressing cells, as actin polymerization everywhere prevents movement in a specific direction.

The strong actin polymerization caused by GefA overexpression was induced by overnight incubation with doxycycline, but the optogenetically recruitable GefA can promote cell expansion within minutes. We were curious to see whether the fast expansion was caused by rapid signaling activity and actin polymerization downstream of Ras activation, independent of transcriptional activities. To address this question, we observed Ras activity and its downstream signaling events, PIP3 accumulation, and actin polymerization directly when recruiting GefA to the cell cortex. Before recruitment, only a single RBD patch appeared at protrusions. Within 15 s after GefA recruitment, the entire membrane displayed RBD binding, indicating a major, global increase in Ras activity. The cell began to expand, making broader protrusions, and the RBD signals remained on the membrane throughout the recruitment period (Fig. 2*F*), even in latrunculin A–treated cells (Fig. 2*G*). The amounts of GefA on the membrane before and after recruitment are shown with line scan (*SI Appendix*, Fig. S2*G*). Although the total amount of recruited GefA on the membrane is not high, it is sufficient to give rise to a 20% increase in Ras activity, as quantified by the normalized cytosolic RBD signals drop (*SI Appendix*, Fig. S2*H*). Concurrently, the LimE biosensor indicated that actin polymerization increased along the cell periphery, consistent with the area expansion of the cell, as quantified in the line scan (Fig. 2*H*). Actin waves were sometimes observed during this expansion, usually emerging from the center and propagating to the edge (Movie S3). Simultaneously, PIP3 accumulation showed a dynamic increase on the membrane. However, the PIP3 response lasted for 5 min before subsiding for the remainder of the time (Fig. 2*I*). The transient nature of the PIP3 response likely explains why we did not observe obvious PIP3 patches on the membrane in cells incubated overnight with doxycycline (Fig. 2*D*).

NGAP Alters Cell Morphology and Migration. Since Ras GTPases are regulated by both GAPs and GEFs (11, 12), we examined the effects of overexpressing or recruiting RasGAPs. Previously, we found that overexpression of a C2 domain-containing RasGAP, C2GAPB, inhibited Ras activity, but when globally expressed, C2GAPB polarized cells and promoted migration (11). Building on this observation, we turned our attention to another C2 domain-containing RasGAP in *Dictyostelium*, NGAP, to investigate its role in cell migration (28).

Given the inhibitory effect of NGAP on Ras activity, we hypothesized that its local recruitment to the leading-edge protrusion would induce protrusion retraction, while its global recruitment would result either in inducing polarity as does C2GAPB or whole-cell contraction and cessation of migration (Fig. 3*A*). To test this hypothesis, we illuminated a small region of the cell with blue light, thereby concentrating SspB-fused NGAP in that region. Within 30 s, the illuminated area of the cell rapidly contracted, causing the cell to start to migrate away from the light source (Fig. 3*B*). Next, we placed cells in microchannel devices. Consistent with the results from the 2D experiments, the enrichment of SspB-fused NGAP at the cell front induced a polarity switch (Fig. 3*C*). NGAP recruitment consistently pushed the cell to move away from the light (Fig. 3*D*). In contrast to cells on 2D substrates, cells in confinement exhibit directed migration, which enables us to alternate their polarity repeatedly using SspB-fused NGAP (Movie S4). This polarity switch is not due to the optogenetics or microchannel system, as the recruitment of empty vector control did not cause a polarity switch (*SI Appendix*, Fig. S3*A*), while the opto-NGAP dramatically changed the cell migration direction, indicated by the negative velocity value (*SI Appendix*, Fig. S3*B*).

We further investigated the effects of overexpressing NGAP on cell morphology and migration. When NGAP expression was induced with doxycycline, cells formed tiny protrusions and rounded up, as compared to the normal-sized pseudopods in control cells with a clear front and back (Fig. 3*E*). To determine whether these tiny protrusions could mediate cell migration, we recorded time-lapsed videos with and without NGAP expressions. While control cells migrated a significant distance over 16 min, NGAP-expressing cells hardly moved (Fig. 3*E*). The migration tracks were captured by color-coded overlay images. Quantification of average cell migration speed showed that with NGAP expression, cell migration speed was 1.5 μm/min, compared to 5 μm/min for control cells (Fig. 3*F*). NGAP-expressing cells also had a smaller aspect ratio but a similar basal cell area and volume when compared to control cells, indicating they were rounder and might

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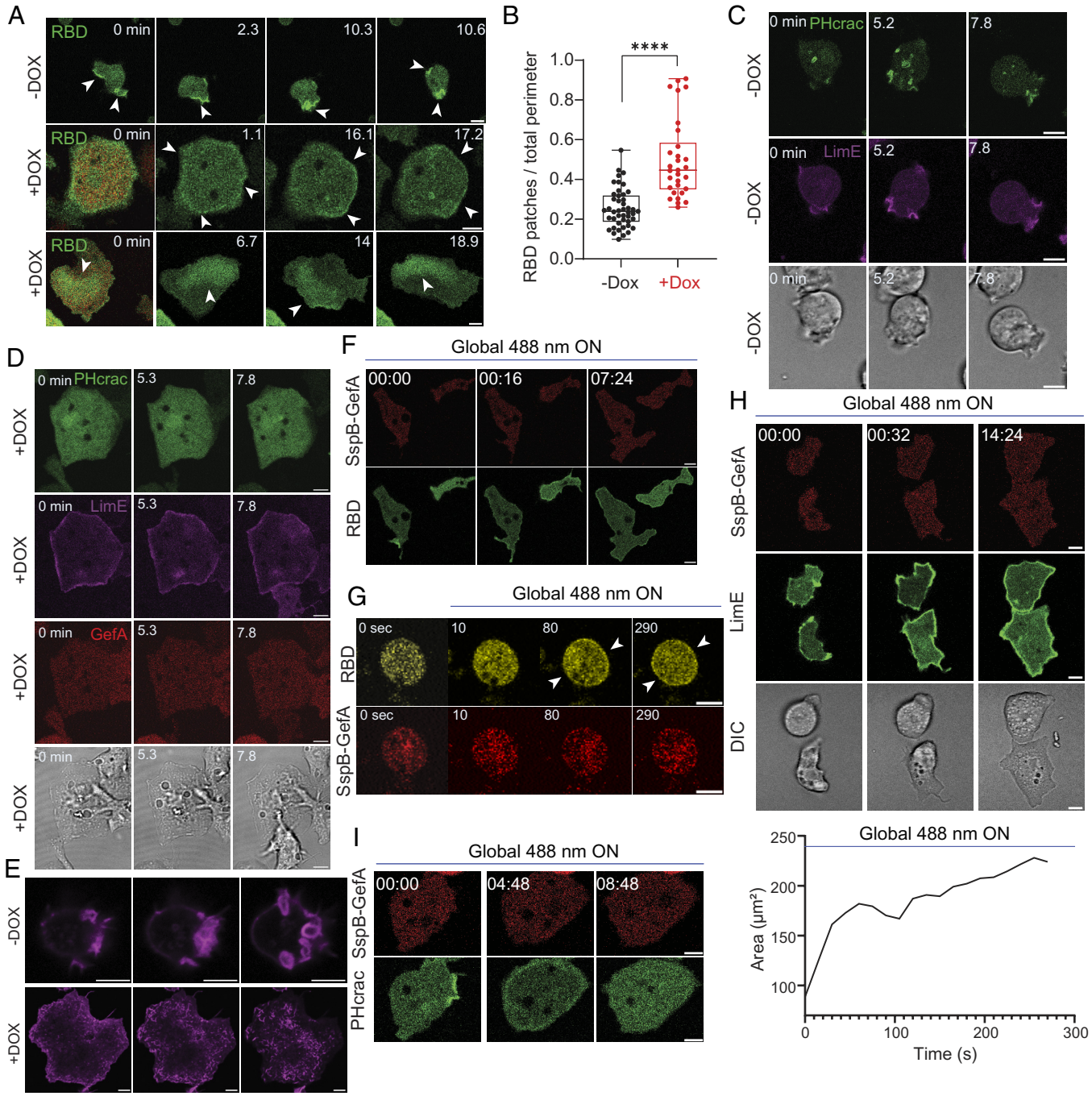

Fig. 2. GefA activates signal transduction and cytoskeletal activities. (*A*) Confocal images of vegetative *Dictyostelium* cells expressing GFP-RBD (biosensor for activated Ras; green) minus (*Top*) and plus (*Middle* and *Bottom*) induction of mRFPmars-GefA (red) expression. White arrows highlight the RBD patches (activities) in these cells. In the *Bottom* panel, RBD shows wave-like activities. Time in minutes. (Scale bars, 5 µm.) (*B*) Box-and-whisker plots of the ratio between RBD patch lengths and total cell perimeter minus (black) or plus (red) GefA expression. n = 45 (−Dox) or n = 30 (+Dox) cells examined over three independent experiments; asterisks indicate significant difference, ****$P \leq 0.0001$ (two-sided Mann–Whitney test). The boxes extend from 25th to 75th percentiles, the median is at the center, and whiskers and outliers are graphed according to Tukey's convention (GraphPad Prism 8). Confocal images of vegetative *Dictyostelium* cells expressing PHcrac-YFP (biosensor for PIP_3; green) and LimE-Halo (biosensor for F-actin; magenta) minus (*C*) and plus (*D*) the induction of mRFPmars-GefA (red) expression. Time in minutes. (Scale bars, 5 µm.) (*E*) Phalloidin staining of vegetative *Dictyostelium* cells minus (*Top*) and plus (*Bottom*) GefA expression, showing in three z-stacks images. Time-lapse confocal images of vegetative *Dictyostelium* cells coexpressing Opto-GefA and RBD-YFP in absence or presence of latrunculin A (*F*–*G*) or LimE-YFP (*H*) or PHcrac-YFP (*I*) on the 2D glass surface. Opto-GefA is recruited globally by switching on a 488 nm laser. The arrow shows membrane RBD patches (*G*). Area changes after recruitment of Opto-GefA in (*H*) are shown in the line plot. Time is in min:s or second *(G)* format. (Scale bars, 5 µm.)

not form large protrusions (Fig. 3 *G* and *H* and *SI Appendix*, Fig. S3*C*). Since NGAP expression reduces cell migration speed, we examined the correlation between NGAP expression levels and migration speed. As might be expected, we detected a negative correlation between NGAP intensity and cell migration speed (*SI Appendix*, Fig. S3*D*), further indicating that NGAP inhibits cell migration.

Next, to directly examine the inhibitory role of NGAP on cell migration and protrusion formation, we globally recruited SspB-fused NGAP to the entire cell cortex. This recruitment of NGAP caused a dramatic whole-cell contraction within a short time (Fig. 3*I* and Movie S5). The cell area also decreased more than twofold after NGAP recruitment (Fig. 3*J*). The cells expressing the recruitable version of NGAP already had impaired cell

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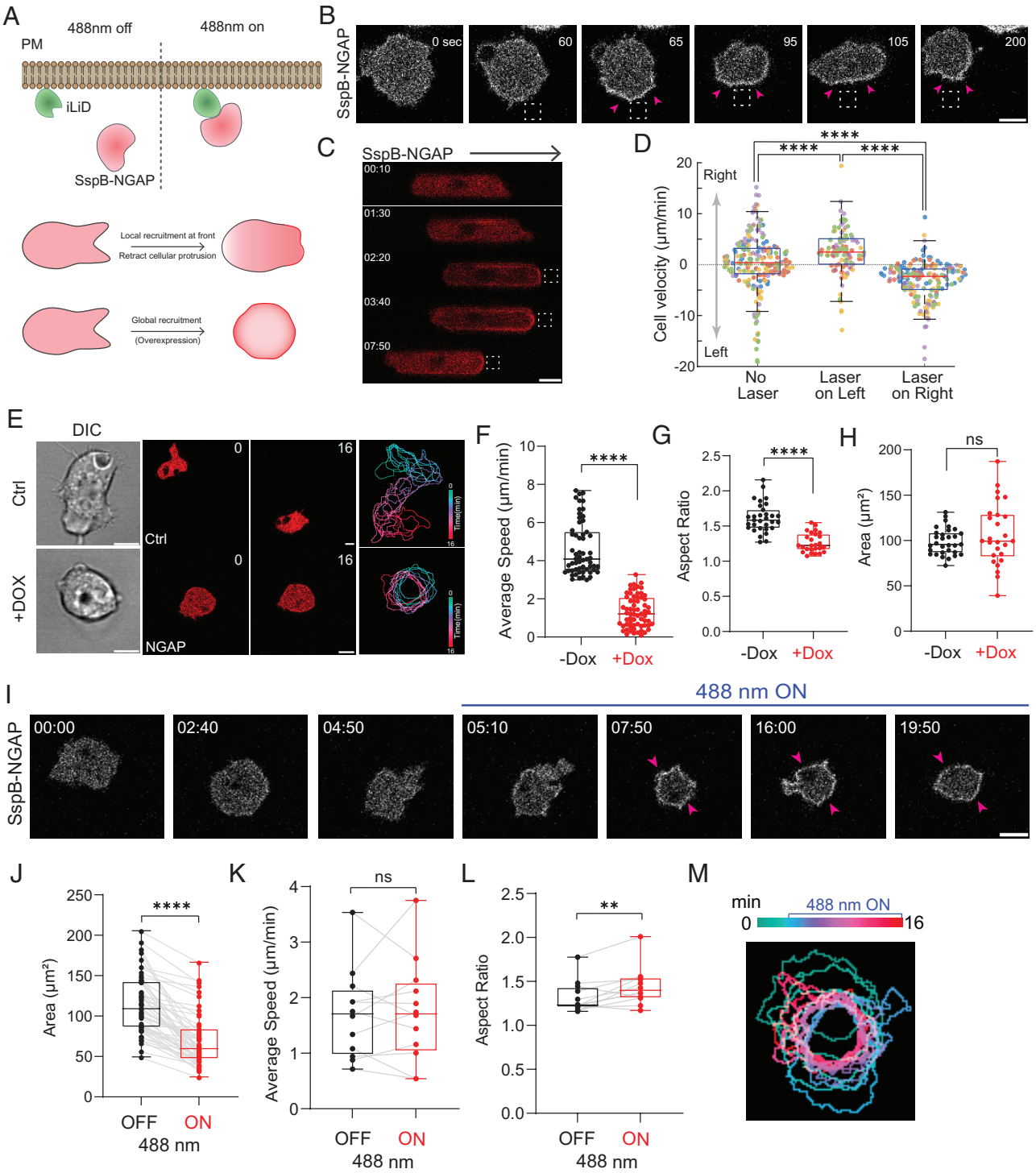

Fig. 3. Inhibition of Ras by NGAP alters cell morphology and migration. (*A*) Schematic of the optogenetics tool. Recruitment of Opto-NGAP (mRFPmars-SspB-NGAP) to membrane-anchored iLiD upon 488 nm illumination. Typical effects at cell front or global recruitment are indicated. (*B*) Opto-NGAP recruited at one side of migrating vegetative *Dictyostelium* cell on the 2D glass surface with 488 nm light (dashed white box). Pink arrows highlight the retraction of the protrusion. (Scale bar, 5 μm.) (*C*) Opto-NGAP recruited to the front of cell migrating to the right in a microchannel (dashed white box). The arrow indicates the migration direction before light stimulus. (Scale bar, 5 μm.) (*D*) Instantaneous velocity of Opto-NGAP cells when the laser light is off, to the left or right of cells. Colored dots represent each of n = 6 cells; total number of measurements are 198, 152, and 152, respectively. Asterisks indicate significant difference, ****$P \leq 0.0001$ (two-sided Mann–Whitney test). (*E*) Time-lapse confocal images of vegetative *Dictyostelium* cells expressing empty vector control (red; *Top*) or mRFPmars-NGAP (red; *Bottom*) after overnight Dox treatment. Time in minutes. (Scale bars, 5 μm.) DIC images (*Left-most* panel) and overlaid outlines (color-coded 1-min intervals) of control or NGAP-expressing cells (*Right-most* panel). Box-and-whisker plots of average cell speed (*F*), Aspect ratio (*G*), and cell area (*H*) before (black; −DOX) or after (red; +DOX) overnight induction of NGAP. n = 31 (−Dox) or n = 27 (+Dox) cells examined over three independent experiments; asterisks indicate significant difference, ****$P \leq 0.0001$; NS, not significant, $P = 0.4888$ (two-sided Mann–Whitney test). The boxes extend from 25th to 75th percentiles, the median is at the center, and whiskers and outliers are graphed according to Tukey's convention (GraphPad Prism 8). (*I*) Snapshots of vegetative *Dictyostelium* cells expressing Opto-NGAP on the 2D glass surface. Opto-NGAP is recruited globally by switching on a 488 nm laser. Pink arrows highlight the contraction of the entire cell. Time in min:s format. (Scale bars, 5 μm.) Box-and-whisker plots of cell area (*J*), average cell speed (*K*), and aspect ratio (*L*) before (black; 488 nm OFF) or after (red; 488 nm ON) global recruitment of NGAP. n = 50 cells (cell area) and n = 12 cells (average speed and aspect ratio) examined over three independent experiments; Statistical analysis as in (*F–H*). NS, not significant, $P = 0.9697$ (two-sided Wilcoxon signed-rank test). (*M*) Color-coded (1-min interval) outlines of a representative cell before and after the presence of 488 nm light.

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migration and rounded morphology before recruitment, and their average migration speed and aspect ratio did not change significantly before and after recruitment (Fig. 3 *K* and *L*). The whole-cell contraction after NGAP recruitment was also shown in the color-coded overlay image (Fig. 3*M*). Based on the local, global recruitment and overexpression results, NGAP appears to inhibit cellular protrusions by inducing contraction, thereby halting cell migration. To further test this idea, we examined the effects of NGAP recruitment in myosin heavy chain null (*mhcA*$^-$) and racE null (*racE*$^-$) cells, both of which have defects in contraction. As expected, neither local nor global recruitment of NGAP caused significant changes in area in the null mutants, compared to the twofold changes in area reduction in the wild type (Fig. 3*I* and *SI Appendix*, Fig. S4 *A*–*H*).

NGAP Alters Signal Transduction and Cytoskeletal Activities. Given that NGAP is a RasGAP, we examined whether the phenotypes caused by its expression or recruitment were due to decreased Ras activity. We first compared RBD signals in cells with or without NGAP expression. Control cells exhibited broad, long-lasting RBD-labeled protrusions, whereas NGAP-expressing cells only had tiny and transient RBD-labeled protrusions. The ratio of RBD patch to cell perimeter length showed that Ras activity decreases with NGAP expression (Fig. 4 *A* and *B*). To validate that this decreased Ras activity is due to the GAP activity of NGAP, we mutated the catalytic arginine R finger (R616) in the GAP domain (28). As expected, the expression of NGAP (R616A) did not affect the size of RBD patches, cells exhibited similar size and intensity of RBD activities with and without NGAP (R616A) expression (*SI Appendix*, Fig. S3*E*). The ratio of RBD patch to cell perimeter also shows no significant difference (*SI Appendix*, Fig. S3*F*), and the expression of this R616A mutant did not slow down the migration of cells (*SI Appendix*, Fig. S3*G*). Similarly, we further assessed PIP3 activity and actin polymerization, which typically mirrors RBD signal patterns. As anticipated, PHcrac and LimE only transiently lit up on small puncta-like protrusions in NGAP-expressing cells, compared to multiple wide patches in control cells, clearly demonstrating that NGAP plays an inhibitory role on Ras, PI3K, and actin polymerization activities (Fig. 4 *C* and *D*). Ras/PIP3 patches are also indicators of macropinocytosis (11); uptake measurements revealed that macropinocytosis was dramatically reduced upon NGAP expression (*SI Appendix*, Fig. S3 *H* and *I*).

Interestingly, among the NGAP-expressing population, some cells are flatter and showed LimE ring patterns around the periphery close to the bottom surface. These actin rings appeared to be "standing waves" that propagated slowly or did not propagate relative to the cell (Fig. 4*E*). This unique ring structure of actin was also confirmed by phalloidin staining in NGAP-expressing cells, which was distinct from the protrusions in control cells (Fig. 4*F*). Both small LimE-labeled protrusions and nonpropagating actin rings could contribute to the decreased migration of NGAP-expressing cells.

We next tested whether NGAP functions as a RasGAP to inhibit Ras activity and its downstream PI3K and actin polymerization activity, thereby slowing down cell migration. To test this, we used our optogenetic system to globally recruit NGAP and monitored RBD and LimE signals as readouts. Before recruitment, we detected tiny RBD and LimE labeling on the small protrusions of NGAP-expressing cells. Upon blue light activation, NGAP was recruited to the cell cortex within 30 s, rapidly diminishing RBD and LimE signals, resulting in whole-cell contraction within 3 min, which was maintained for at least 10 min (Fig. 4 *G* and *H* and Movie S6). Altogether, these results indicate that NGAP inhibits cell migration by decreasing Ras activity and actin polymerization, leading to the elimination of protrusions and cell contraction.

Optimum Ras Activity and Cortical Tension Dictate Effective Migration. Expression of three Ras regulators, GefA, C2GAPB, and NGAP, cause either spreading, tightly localized protrusions, or tiny unfocused protrusions, which either halted migration, increased migration, or decreased migration, respectively. The shape and size of the protrusions also suggested that differences in mechanical properties, such as cortical tension, may occur.

To measure their cortical tension, we performed micropipette aspiration (MPA) in GefA-, C2GAPB-, and NGAP-expressing cell lines and compared the values to those of wild-type cells (38, 39) (Fig. 5*A*). We determined the length of extension into the pipette versus the cell radius at various pressures and calculated the cortical tension (Fig. 5*B*). We observed that NGAP- and C2GAPB-expressing cells had lower cortical tension than the control cells, while GefA-expressing cells had increased cortical tension (Fig. 5*C*).

These initial cortical tension measurements were conducted on cell lines that stably express the Ras regulators. In contrast, the signaling and morphological changes observed in fast-moving cells are much more rapid. To test whether Ras manipulation has immediate effects on cortical tension, we leveraged our optogenetic system. Because the light-induced recruitment occurred immediately and is reversible, we decided to measure the cortical tension of the same cell twice. The first time was performed without blue light illumination. After the cell reached the critical point, we relaxed the cell and then measured the cortical tension again with blue light activation. We hypothesized that acutely altered Ras activity would augment the change of cortical tension, and as expected, recruitment of GefA further increased cortical tension by about 1.7-fold while recruitment of NGAP decreased cortical tension ~40% from 0.8 nN/μm to 0.5 nN/μm (Fig. 5 *D*–*G*). However, the recruitment of empty vector control had little effect on the change of cortical tension before or after recruitment (Fig. 5 *J* and *K*). Because recruitment of C2GAPB increased myosin II contraction, it might lead to an increase in the cortical tension. As shown in Fig. 5 *H* and *I*, recruitment of C2GAPB slightly increased the cortical tension, but not as significantly as the changes elicited by recruiting GefA or NGAP, which could mean that C2GAPB is within the optimal range of both Ras activity and cortical tension for most effective migration.

Discussion

In this study, we expanded our investigation into the role of Ras activity in regulating cellular migration dynamics. Using *Dictyostelium* as a model, we demonstrated that acute modulation of Ras activity via Ras GEFs and GAPs induces striking alterations in cell shape and migration behavior. By employing optogenetic tools, we replicated phenotypes previously observed in overexpression studies, but on significantly shorter timescales, highlighting the immediate influence of Ras on cell motility. Furthermore, the immediate effects of Ras modulation are independent of receptor inputs and extracellular gradients, underscoring the self-autonomously organized properties of the signaling network. Additionally, we examined how changes in Ras activity affect the mechanical properties of the cell, revealing dynamic fluctuations in cortical tension that correspond with variations in Ras signaling. Our findings establish a biphasic relationship between Ras activity levels, cortical tension, and migration efficiency, emphasizing the necessity of balanced Ras signaling and cortical tension for optimal cellular motility.

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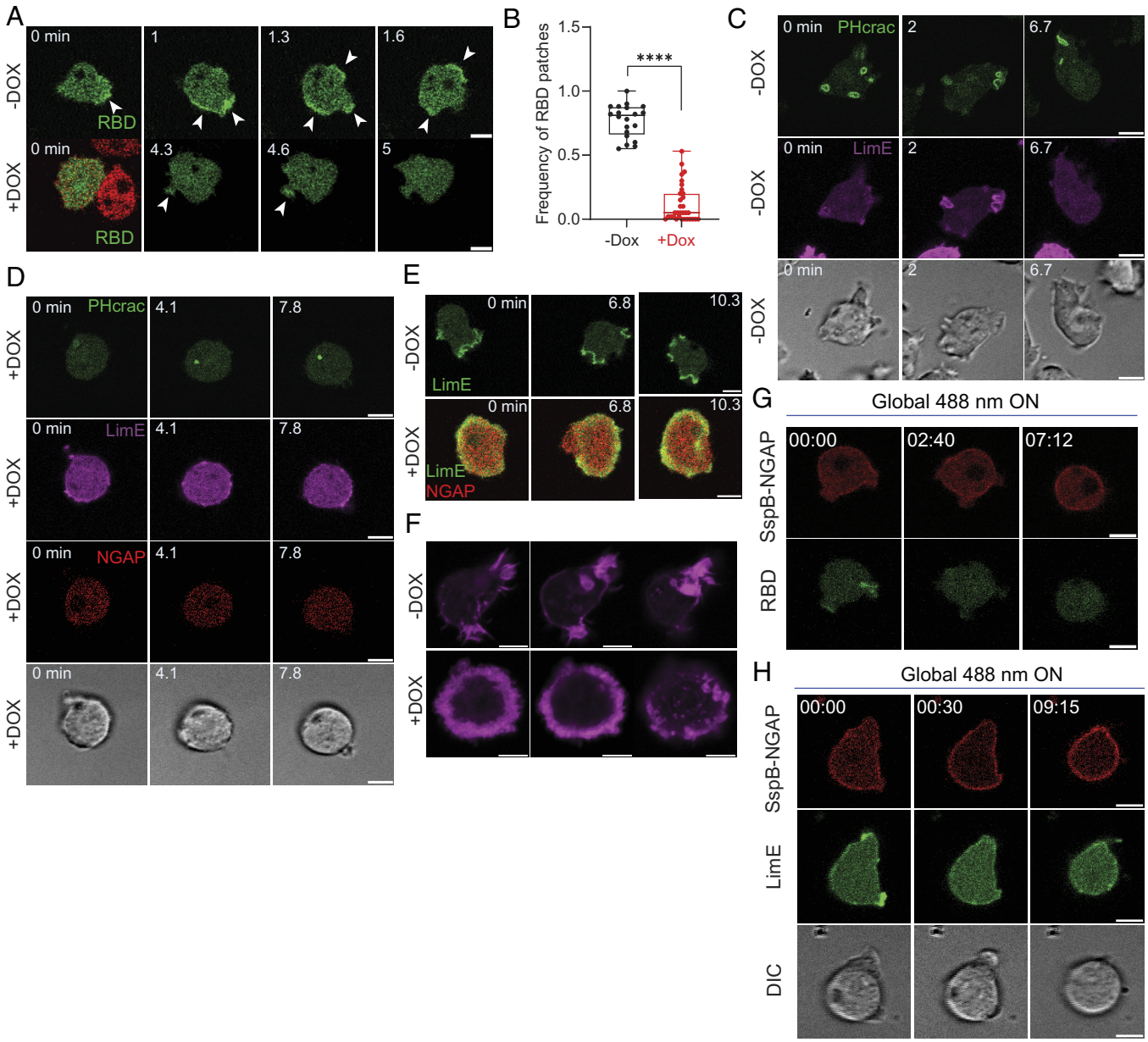

Fig. 4. NGAP alters signal transduction and cytoskeletal activities. (*A*) Confocal images of vegetative *Dictyostelium* cells expressing GFP-RBD (biosensor for activated Ras; green) minus (*Top*) and plus (*Bottom*) induction of mRFPmars-NGAP (red) expression. White arrows highlight the RBD patches (activities) in these cells. (*B*) Box-and-whisker plots of the ratio between RBD patch lengths and total cell perimeter minus (black) or plus (red) NGAP expression. n = 20 (−Dox) or n = 32 (+Dox) cells examined over three independent experiments; asterisks indicate significant difference, ****$P \leq 0.0001$ (two-sided Mann–Whitney test). The boxes extend from 25th to 75th percentiles, the median is at the center, and whiskers and outliers are graphed according to Tukey's convention (GraphPad Prism 8). Confocal images of vegetative *Dictyostelium* single cells expressing PHcrac-YFP (biosensor for PIP3; green) and LimE-Halo (biosensor for F-actin; magenta) minus (*C*) and plus (*D*) induction of mRFPmars-NGAP (red) expression. Time in minutes. (Scale bars, 5 μm.) (*E*) Time-lapse confocal images of vegetative *Dictyostelium* cells expressing LimE-GFP minus (*Top*) and plus (*Bottom*) NGAP expression. Actin ring is visible at the bottom surface of NGAP-expressing cells. Time in minutes. (Scale bars, 5 μm.) (*F*) Phalloidin staining of vegetative *Dictyostelium* cells minus (*Top*) or plus (*Bottom*) NGAP expression, showing in three z-stacks images. Time-lapse confocal images of vegetative *Dictyostelium* cells coexpressing Opto-NGAP and RBD-YFP (*G*) or LimE-YFP (*H*) on the 2D glass surface. Opto-NGAP is recruited globally by switching on a 488 nm laser. Time is shown in min:s format. (Scale bars, 5 μm.)

Distinct outcomes were observed between local and global recruitment of RasGEF and RasGAP. While local recruitment promoted or inhibited protrusions as anticipated, global recruitment of these Ras modulators resulted in immobilized cell movement. We propose that the amplification of Ras signaling by global recruitment pushes the system to an extreme imbalance, which results in reduction in cell motility. In our previous work, we identified an optimal level of Ras activity that maximizes migration in several cellular systems (11). Building on this, our current findings further suggest a biphasic relationship between Ras activity and cell migration, where both insufficient and excessive Ras activity impair motility. Furthermore, as cortical tension has been implicated in regulating cell shape and migration behavior (40–43), we characterized the effects of our Ras modulators on cortical tension. The results showed that cortical tension parallels change in Ras activity. Together, these findings indicate the existence of a "Goldilocks," optimal range of Ras activity and cortical tension that regulates cell behavior, as demonstrated in this study through its impact on migration.

On an unrestricted 2D glass surface, while local recruitment of RasGEF or RasGAP did induce or inhibit protrusions, cells lacked sufficient polarity to support sustained changes in cell migration. In contrast, within uncoated channels, cells migrated as polarized cells even without any external stimulus. The channel setup also allowed for more precise optogenetic manipulation of migration

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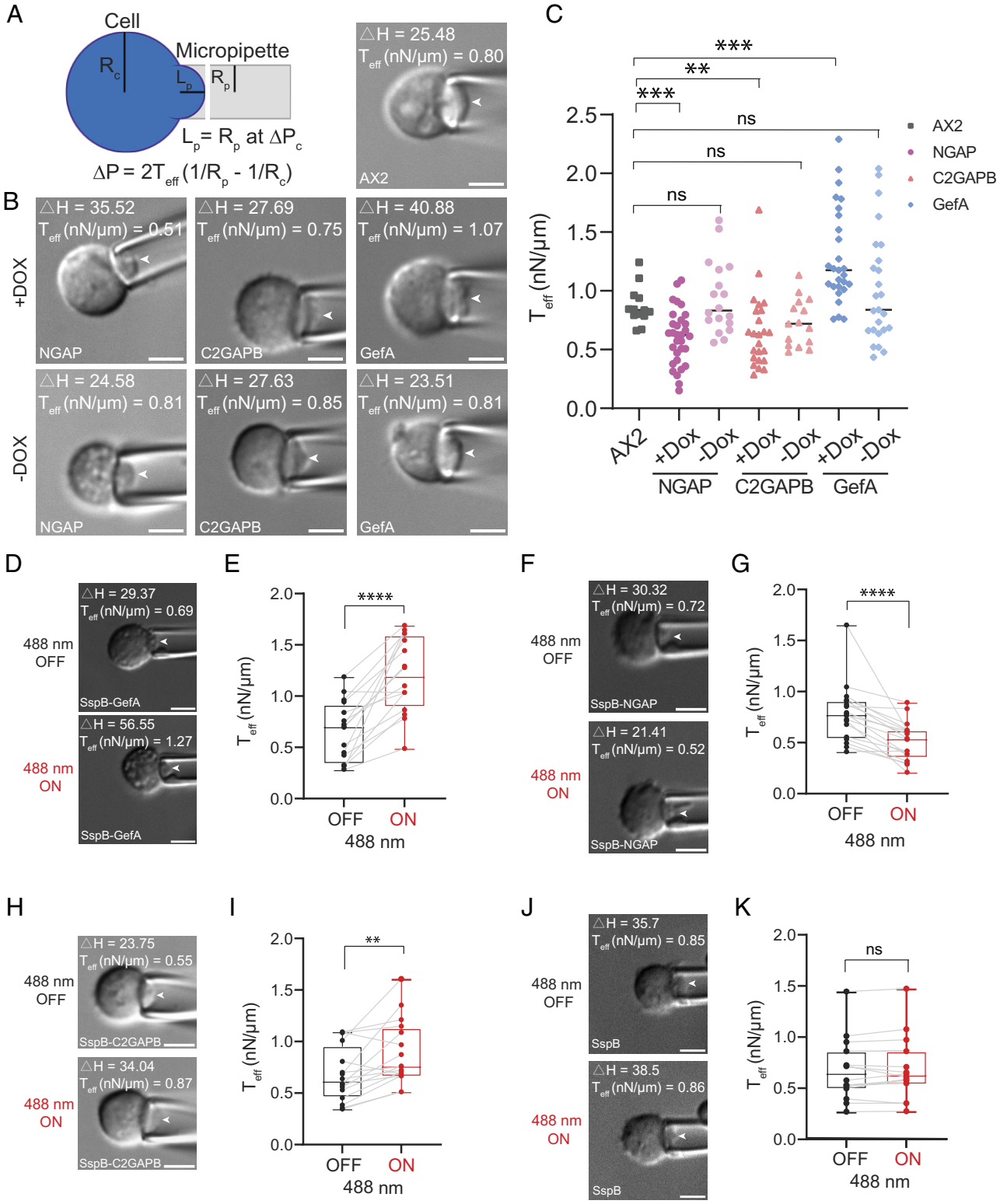

Fig. 5. Modulation of Ras activity leads to changes in cortical tension. (*A*) Schematic of the MPA and example of MPA measurement on the wild-type cell (AX2, *Right*). For cortical tension measurements, the pressure increases until the length (L_p) of the region of the cell pulled into the micropipette equals the radius of the pipette (R_p). At this equilibrium pressure (ΔP_c), the radius of the cell can be measured, and the effective cortical tension T_{eff} is calculated from the equation shown in the panel. (*B*) Representative images of MPA measurements on NGAP, C2GAPB, and GefA-expressing cells plus (*Top*) and minus (*Bottom*) the addition of DOX. (*C*) The effective cortical tension values of AX2, NGAP (±Dox), C2GAPB (±Dox), and GefA (±Dox). Measurements were quantified as detailed in the *Materials and Methods*. ***$P \leq 0.001$; **$P \leq 0.01$; NS, not significant, $P = 0.9528$, $P = 0.1077$, $P = 0.7847$. Cell numbers n = 13 to 27 from three independent experiments. *P* values were calculated using Kruskal–Wallis ($P \leq 0.0001$) followed by the two-sided Mann–Whitney test, GraphPad Prism 8. Representative images and effective cortical tension values of MPA measurements on SspB-GefA *(D–E)*, SspB-NGAP *(F–G)*, SspB-C2GAPB *(H–I)*, and SspB-Ctrl *(J–K)* cells before (*Top*, 488 nm OFF) and after (*Bottom*, 488 nm ON) recruitment. (Scale bars, 5 μm.) Measurements were quantified as detailed in the *Materials and Methods*. ****$P \leq 0.0001$; **$P = 0.0027$; NS, not significant, $P = 0.2979$. Cell numbers n = 16 to 18 from three independent experiments. *P* values were calculated using the two-sided Mann–Whitney test, GraphPad Prism 8.

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direction and behavior. In the channels, it was clear that local enrichment of RasGEF at the trailing edge of migrating cells in confinement was sufficient to reverse their migration directionality. Similarly, optogenetic recruitment of RasGAP, albeit at the leading edge of migrating cells, changed migration direction. The effects of these distinct optogenetic stimulations on migration directionality were evident in microchannels than on 2D because of the highly directed migration induced by confining microchannel walls. Importantly, the accumulation of RasGEF or RasGAP consistently preceded the directional changes of the cells. Furthermore, even on the 2D unrestricted surface, the global activation of Ras through RasGEF recruitment overrode chemotactic stimuli in wild-type cells, and local activation directly induced protrusions and polarity changes in Gβ-null cells. These findings provide strong evidence that Ras activity provides a powerful signaling input necessary for directed migration, functioning independently of receptor-mediated input.

Although Ras is generally believed to regulate cellular behavior and cytoskeletal remodeling through long-term gene regulation (44–46), our previous work and this study using optogenetics demonstrate that Ras controls cell migration over short time periods (11, 12). Notably, both cell shape and migration behaviors changed dramatically within seconds to minutes, accompanied by corresponding changes in cortical tension as Ras activity was modulated. Although cell state changes are often associated with transcriptional regulation (47), our findings highlight the effects of short-term Ras activity fluctuations that are independent of transcriptional changes. Our results indicate that these rapid, nontranscriptional mechanisms can acutely shift the cell state, potentially by driving the system away from its current setpoint. While transcriptional regulation may ultimately establish a new setpoint for long-term adaptation, this raises the question of how these immediate, nontranscriptional effects integrate and coordinate with longer-term transcriptional and translational changes to influence overall cell behavior.

Our findings have important implications for cancer treatment, suggesting that investigators targeting Ras should consider the cell's activity state and its microenvironment. Given the biphasic relationship between Ras activity and cell migration efficiency, an ideal therapeutic approach for Ras-driven oncogenesis would aim to shift the activity level beyond the optimal range for migration, driving it toward an inactive or highly active state (48–50). A deeper understanding of the distinct roles Ras plays in regulating cell migration, mechanical properties, and growth is crucial for the development of more targeted and effective therapeutic approaches.

Materials and Methods

For all experiments, *D. discoideum* AX2 and mutant strains were cultured in HL5 medium (11). Microfluidics devices were fabricated as previously described (51). MPA and effective cortical tension quantification were described in the previous review (39). A detailed description of materials and methods is provided in *SI Appendix, SI Materials and Methods*.

Data, Materials, and Software Availability. All study data are included in the article and/or supporting information.

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